

NAME: R.JAGAN

REG NO:192321102

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CO3 - AT2 - CASE STUDY

1..Real-Time Anomaly Detection in Smart Healthcare using Bayesian Belief Networks

1.1 Problem Context

A smart healthcare monitoring system continuously collects physiological data — heart rate (HR), blood pressure (BP), oxygen saturation (SpO2), respiratory rate (RR), and body temperature — from wearable sensors attached to a patient. The goal is to detect abnormal patterns (anomalies) that may indicate a medical emergency such as arrhythmia, hypoxia, or sepsis, while accounting for the fact that vital signs are not independent of one another and that sensor readings are inherently noisy. A Bayesian Belief Network (BBN), also called a Bayesian Network, is well suited to this problem because it represents a joint probability distribution over a set of correlated random variables using a directed acyclic graph (DAG), and it naturally supports probabilistic reasoning under uncertainty.

1.2 Bayesian Belief Network Model Design

The BBN is constructed in two stages: (i) structure learning — determining which nodes (variables) are connected and in what causal/dependency direction, and (ii) parameter learning — estimating the conditional probability tables (CPTs) for each node given its parents, typically from historical patient data using Maximum Likelihood Estimation or Bayesian parameter estimation with Dirichlet priors.

Network Nodes

- Root/latent node: Patient Health State (Normal, At-Risk, Critical)
- Observable child nodes: Heart Rate, Blood Pressure, SpO2, Respiratory Rate, Body Temperature
- Contextual nodes: Patient Age Group, Activity Level (rest/exercise), Sensor Reliability
- Output node: Anomaly Alert (No Alert, Warning, Critical Alert)

Joint Probability Factorisation

For the network above, the joint distribution over all n variables factorises using the chain rule of Bayesian networks as:

$$P(X_1, X_2, \dots, X_n) = \prod_{i=1}^n P(X_i \mid \text{Parents}(X_i))$$

For example, if Anomaly Alert (A) depends on Health State (H), and Health State depends on Activity Level (L) and Age Group (G), while HR, BP, SpO2 and RR each depend on Health State and Activity Level, the joint probability can be written as:

$$P(H,L,G,HR,BP,SpO2,RR,A) = P(G) \cdot P(L) \cdot P(H|L,G) \cdot P(HR|H,L) \cdot P(BP|H,L) \cdot P(SpO2|H,L) \cdot P(RR|H,L) \cdot P(A|H)$$

This factorisation drastically reduces the number of parameters that must be estimated compared to a full joint table, which would require an exponential number of entries; the BBN instead requires only local conditional probability tables sized according to each node's parents.

1.3 Modeling Conditional Dependencies Between Physiological Parameters

Vital signs are physiologically coupled — for instance, a drop in SpO2 is frequently accompanied by a compensatory rise in HR and RR, and BP tends to fall during severe hypoxia. Representing these signals as independent variables (as a naïve threshold-based alarm system would) leads to a high false-alarm rate because natural covariation is misclassified as anomalous. The BBN instead encodes these dependencies explicitly as edges with conditional probability tables, so that the network can distinguish a physiologically consistent deviation (e.g., HR rising together with RR during exercise) from an inconsistent one (e.g., HR rising while BP and SpO2 fall simultaneously at rest, which signals possible shock).

Two categories of dependency are modelled:

- Causal dependency — the latent Health State node is the common cause of correlated changes across HR, BP, SpO2 and RR; conditioning on Health State makes the vital signs conditionally independent of each other (the standard BBN assumption).
- Contextual dependency — Activity Level and Age Group act as evidence nodes that shift the expected baseline distribution of each vital sign, preventing the system from treating an elevated HR during exercise as identical to an elevated HR at rest.

During inference, when a new set of sensor readings arrives as evidence, the network performs belief updating using Bayes' theorem to compute the posterior probability of each possible Health State:

$$P(H | HR,BP,SpO2,RR) = [P(HR,BP,SpO2,RR | H) \cdot P(H)] / P(HR,BP,SpO2,RR)$$

Exact inference is carried out using variable elimination or junction-tree propagation for small networks; for larger or dynamic networks, approximate inference (Markov Chain Monte Carlo sampling or loopy belief propagation) is used to keep computation tractable in real time.

1.4 Handling Uncertainty Due to Sensor Noise

Wearable sensors are subject to motion artefacts, battery-related drift, poor skin contact, and transmission dropouts, all of which introduce noise into the observed readings. The BBN handles this uncertainty in several complementary ways:

- Soft evidence instead of hard evidence: rather than asserting a sensor reading as an absolute fact, the system treats it as a likelihood distribution (virtual evidence), so a noisy reading only partially shifts the belief in the underlying Health State rather than overriding it.
- Sensor Reliability node: an explicit node models the probability that a given sensor is functioning correctly; when reliability is estimated to be low (e.g., due to detected signal clipping or an implausible rate of change), the influence of that sensor's reading on the posterior is automatically down-weighted.
- Gaussian/discretised noise models in the CPTs: continuous vital-sign readings are modelled with a Gaussian conditional density (or discretised into bins) around the expected mean for each

Health State, so a single noisy outlier has a bounded effect on the posterior rather than triggering an immediate false alarm.

- Temporal smoothing via Dynamic Bayesian Networks (DBN): the BBN is extended across time slices, so the belief in the current Health State is combined with the belief from the previous time step (a Hidden Markov-style transition model), which suppresses transient sensor spikes that do not persist.

1.5 Probability Learning (Parameter Estimation)

The conditional probability tables are learned from labelled historical ICU/wearable datasets using:

1. Maximum Likelihood Estimation (MLE) when sufficient labelled data is available for each parent-configuration.
2. Bayesian estimation with Dirichlet priors when data is sparse for some configurations, which prevents zero-probability entries and incorporates clinical prior knowledge (e.g., known normal ranges for HR and SpO2).
3. Online/incremental updating, where the CPTs are periodically refined as new confirmed-diagnosis data arrives, allowing the model to adapt to a specific patient's personal baseline over time (personalised priors).

1.6 System Performance Evaluation

The anomaly-detection performance of the BBN is typically evaluated on a held-out set of labelled patient-monitoring episodes using standard classification metrics:

Metric	Definition	Typical Target in Healthcare Monitoring
Sensitivity (Recall)	Proportion of true critical events correctly flagged	> 95% (missed critical events are costly)
Specificity	Proportion of true normal states correctly left un-flagged	> 90% (to control alarm fatigue)
Precision	Proportion of raised alerts that are true anomalies	As high as possible without hurting recall
F1-Score	Harmonic mean of precision and recall	Used to balance the two
AUC-ROC	Discrimination ability across alert thresholds	> 0.9 considered strong
Alarm rate / hour	Operational measure of alert burden on clinicians	Minimised subject to sensitivity constraint

In practice, BBN-based systems report a marked reduction in false alarms compared to simple threshold rules, because the network jointly reasons over correlated vitals instead of firing on any single out-of-range reading; however, sensitivity gains typically require carefully elicited priors and sufficiently rich training data across diverse patient sub-populations (age, comorbidities), and performance should always be reported separately for these sub-groups to check for bias.

1.7 Challenges in Real-Time Deployment

- Computational latency: exact inference can become intractable as the network grows (more sensors, more contextual nodes); approximate/streaming inference is required to meet real-time alerting deadlines.
- Missing and asynchronous data: wearables sample at different rates and occasionally drop packets, requiring the network to perform inference gracefully with partial evidence.
- Concept drift: a patient's physiological baseline changes with age, medication, or disease progression, so static CPTs become stale and must be periodically relearned.
- Cold-start problem: newly onboarded patients have no personal history, forcing reliance on population-level priors that may not fit them well.
- Privacy and data governance: continuous physiological data is highly sensitive, requiring on-device or edge inference and strict access control.
- Alarm fatigue: even a modest false-positive rate becomes clinically disruptive at scale, so the alerting threshold must be tuned against clinician workload, not accuracy alone.
- Explainability: clinicians require the network's reasoning (which parents contributed to a Critical Alert) to be inspectable, which favours BBNs over black-box models for this domain.

1.8 Conclusion

Bayesian Belief Networks provide a principled framework for smart healthcare monitoring because they explicitly encode the conditional dependencies among physiological signals through a compact DAG structure, propagate sensor uncertainty as probability distributions rather than hard thresholds, and support transparent, explainable inference of a patient's underlying health state. When combined with dynamic (time-sliced) extensions and continuously updated probability learning, such systems can substantially reduce false alarms while preserving high sensitivity to genuine emergencies, although latency, data quality, drift and privacy constraints remain significant engineering challenges for real-time deployment at scale.

2. Probabilistic Learning for Network Intrusion Detection

2.1 Problem Context

A network intrusion detection system (NIDS) must classify streams of network traffic (features such as packet size, protocol type, connection duration, number of failed logins, byte counts, and flag combinations) as normal or malicious in real time. Because attack strategies evolve and complete labelled data is rarely available, the system combines three complementary probabilistic learning techniques: the Naïve Bayes classifier for fast supervised classification, the Expectation-Maximisation (EM) algorithm for learning from unlabelled or partially labelled traffic, and Bayesian learning for continuously updating beliefs about model parameters as new evidence arrives.

2.2 Naïve Bayes Classifier for Traffic Classification

The Naïve Bayes classifier assumes that, given the class label C (Normal or Attack, or a specific attack category such as DoS, Probe, R2L, U2R), the individual traffic features $F_1, F_2, \dots, F_n$ are conditionally independent. The posterior probability of a class given an observed feature vector is computed using Bayes' theorem:

$$P(C | F_1, \dots, F_n) = [P(C) \cdot \prod_{i=1..n} P(F_i | C)] / P(F_1, \dots, F_n)$$

Since the denominator is constant across classes, classification reduces to selecting the class that maximises the numerator:

$$\hat{C} = \operatorname{argmax}_C P(C) \cdot \prod_{i=1..n} P(F_i | C)$$

Continuous features (e.g., connection duration, byte counts) are modelled with a Gaussian likelihood, while categorical features (e.g., protocol type, service, flag) use multinomial probability tables estimated by frequency counting with Laplace (add-one) smoothing to avoid zero-probability issues for rare but legitimate feature values. The independence assumption, although technically violated by real network features, makes the classifier extremely fast — a critical property for line-rate intrusion detection — and it remains competitive in practice because errors introduced by the independence assumption are often correlated across classes and therefore partly cancel out in the argmax decision.

2.3 EM Algorithm for Hidden Patterns and Incomplete Data

Labelling network traffic as attack/normal at scale is expensive, and many collected sessions are unlabelled or only partially labelled; furthermore, attacks often contain hidden sub-types that are not explicitly annotated (e.g., a 'Probe' label might silently contain several distinct scanning techniques). The EM algorithm is used to learn a mixture model over the traffic features in which the true class or sub-cluster is treated as a latent (hidden) variable Z , iterating between two steps until convergence:

- E-step (Expectation): given the current parameter estimates θ , compute the posterior (responsibility) of each hidden class for every unlabelled sample: $\gamma(z_{ik}) = P(Z_i = k | F_i, \theta)$.
- M-step (Maximisation): re-estimate the parameters θ (class priors, means/variances or multinomial probabilities of each feature per class) that maximise the expected complete-data log-likelihood, weighting each sample by its responsibility from the E-step.

$$\theta(t+1) = \operatorname{argmax}_{\theta} \sum_i \sum_k \gamma(z_{ik}) \cdot \log[P(F_i | Z_k, \theta) \cdot P(Z_k)]$$

This process is repeated until the log-likelihood converges, producing both refined estimates of the Naïve Bayes parameters and a soft clustering of traffic into behavioural groups — effectively

discovering hidden attack sub-patterns and filling in the missing labels for unlabelled sessions. EM therefore allows the NIDS to be trained in a semi-supervised manner: a small amount of confidently labelled traffic seeds the initial parameters, and the much larger pool of unlabelled traffic is incorporated through the E/M iterations rather than discarded.

2.4 Bayesian Learning for Continuous Model Updating

Because attacker behaviour is non-stationary, the system treats the Naïve Bayes and mixture-model parameters themselves as random variables with a prior distribution, and updates this distribution as new (possibly delayed) ground-truth labels arrive — e.g., analyst-confirmed incidents. Using Bayes' rule at the parameter level:

$$P(\theta \mid D) \propto P(D \mid \theta) \cdot P(\theta)$$

A Dirichlet prior is a natural conjugate choice for the categorical feature probabilities and class priors, and a Normal-Inverse-Gamma prior is used for Gaussian-modelled continuous features, because conjugacy allows the posterior to be updated in closed form as each new labelled batch D arrives, without retraining from scratch. This Bayesian (rather than purely frequentist) updating gives the system a principled way to represent its own confidence — parameters based on little evidence retain wide posterior uncertainty, so the model naturally relies more heavily on the prior (older, well-established behaviour patterns) until enough new evidence accumulates to shift its beliefs.

2.5 Handling Hidden Patterns and Incomplete Data — Summary of Interactions

Technique	Role	Addresses
Naïve Bayes	Fast supervised classification of traffic as Normal/Attack	Real-time decision-making at line rate
EM Algorithm	Discovers latent attack sub-types; imputes missing labels	Hidden patterns and incomplete/unlabelled data
Bayesian Learning	Continuously updates class and feature probabilities as evidence arrives	Concept drift and evolving attacker behaviour

2.6 Evaluating Effectiveness — Reducing False Alarms and Improving Accuracy

NIDS effectiveness is evaluated using a confusion matrix over Normal/Attack (or multi-class attack categories), from which the following are derived:

Metric	Formula	Operational Significance
Detection Rate (Recall)	TP / (TP + FN)	Fraction of real attacks caught
False Positive Rate	FP / (FP + TN)	Directly drives analyst alert fatigue
Precision	TP / (TP + FP)	Trustworthiness of a raised alert
F1-Score	$2 \cdot \text{Precision} \cdot \text{Recall} / (\text{Precision} + \text{Recall})$	Balanced summary measure
ROC-AUC	Area under TPR vs FPR curve	Threshold-independent discrimination

Empirically, on benchmark intrusion datasets, pure threshold or signature-based systems tend to produce a high false-positive rate against novel or obfuscated attacks, whereas hybrid Naïve Bayes + EM + Bayesian-updating pipelines lower the false-positive rate over time because: (a) EM-derived sub-clusters allow finer-grained normal-behaviour modelling instead of one broad 'normal' class, which reduces the chance that an unusual-but-legitimate session is flagged; and (b) Bayesian posterior updating lets the false-alarm-prone parameters shrink in influence as more confirmed-normal evidence accumulates, rather than requiring a full batch retrain. The trade-off is that EM introduces a risk of converging to a local optimum (mislabelling a genuine attack cluster as a normal sub-cluster), so the system typically constrains EM with a small amount of trusted labelled 'seed' data and validates cluster purity against analyst feedback.

2.7 Effectiveness in Dynamic Environments

- **Adaptability:** Bayesian updating and periodic EM re-clustering let the model track gradually evolving traffic patterns without full retraining downtime.
- **Zero-day resilience:** because the model reasons probabilistically over feature distributions rather than fixed signatures, moderately novel attacks that shift known feature distributions can still raise elevated posterior 'attack' probability.
- **Scalability:** Naïve Bayes' linear-time scoring keeps per-packet/per-flow classification cost low even as traffic volume grows.
- **Limitations:** strong feature independence assumptions can be violated by correlated attack signatures (e.g., coordinated multi-stage attacks), and EM can be sensitive to initialisation and the assumed number of latent clusters, both of which motivate hybridising with ensemble or deep-learning components in high-stakes deployments.

2.8 Conclusion

Combining Naïve Bayes, the EM algorithm, and Bayesian learning gives an intrusion detection pipeline that is fast enough for real-time use, capable of exploiting large volumes of unlabelled traffic to discover hidden attack sub-patterns, and able to continuously refine its beliefs as new evidence and analyst feedback arrive. This probabilistic approach measurably reduces false alarms relative to static rule-based systems while maintaining strong detection accuracy, though careful handling of the independence assumption, EM initialisation, and prior selection is required for robust performance in dynamic, adversarial network environments.

3. Bayesian Inference for Crop Yield Prediction Under Environmental Uncertainty

3.1 Problem Context

Crop yield depends on a combination of environmental factors — rainfall, soil quality (nutrient content, pH, moisture-holding capacity), temperature, humidity, and management practices (irrigation, fertiliser use) — many of which are uncertain at the time a prediction is needed (e.g., at planting time, seasonal rainfall for the rest of the season is not yet known). Bayesian inference is well suited to this problem because it allows prior agricultural knowledge (historical yield patterns, agronomic expertise) to be formally combined with observed seasonal data, and it naturally produces a probability distribution over predicted yield rather than a single deterministic number, giving farmers and planners an explicit measure of confidence.

3.2 Predictive Model Design

The system is modelled as a Bayesian Network (or, at the parameter level, Bayesian linear/regression inference) in which Yield (Y) is the target node with parent nodes representing the key environmental and agronomic drivers:

- Rainfall (R) — seasonal cumulative rainfall, often discretised into Deficient / Normal / Surplus
- Soil Quality (S) — composite index of nutrient level, pH suitability, and organic matter
- Temperature (T) — average and extreme temperature during critical growth stages
- Crop Variety / Management (M) — seed variety, irrigation, fertiliser regime
- Yield (Y) — child node, the quantity to be predicted (e.g., tonnes/hectare, discretised into Low / Medium / High or modelled continuously)

Using the Bayesian Network factorisation, the joint distribution is:

$$P(R, S, T, M, Y) = P(R) \cdot P(S) \cdot P(T) \cdot P(M) \cdot P(Y | R, S, T, M)$$

At prediction time, the system observes partial evidence — e.g., soil-test results and early-season rainfall — and computes the posterior distribution over yield by Bayesian updating:

$$P(Y | R, S, T, M) = [P(R, S, T, M | Y) \cdot P(Y)] / P(R, S, T, M)$$

Where the full parameter set is continuous, a Bayesian (probabilistic) regression formulation is used instead, in which yield is modelled as $Y = \beta_0 + \beta_1 R + \beta_2 S + \beta_3 T + \beta_4 M + \epsilon$, and Bayesian inference is applied not just to Y but to the regression coefficients β themselves, giving a full posterior distribution $p(\beta | \text{data})$ rather than single point estimates, from which the predictive distribution of yield for a new season is obtained by integrating over parameter uncertainty:

$$P(Y_{\text{new}} | D) = \int P(Y_{\text{new}} | \beta) \cdot P(\beta | D) d\beta$$

3.3 Role of Prior Agricultural Knowledge

Priors $P(\beta)$ or $P(Y)$ are elicited from two complementary sources:

- Historical district/region-level yield records over multiple seasons, which give an empirical prior distribution reflecting typical yield ranges under known rainfall/soil regimes.
- Agronomic domain expertise, encoded as informative priors — for instance, agronomists know that yield response to rainfall is nonlinear and saturates beyond a certain threshold, or that a particular soil pH range is essential for a given crop; this knowledge can be encoded directly into

the prior's functional form or into the network's conditional probability tables, rather than requiring the model to rediscover it purely from limited data.

This is particularly valuable in agriculture because the number of full growing seasons of data for a given field is small (often only a few decades at most, and often far fewer per district), so a purely data-driven (maximum-likelihood) model would be highly unstable; the Bayesian prior regularises the estimate and prevents overfitting to a handful of unusual seasons (e.g., one extreme drought year).

3.4 Modeling Seasonal Variation

Seasonal variation is incorporated in two ways:

- Hierarchical/dynamic priors: the prior for the current season's parameters is set from the posterior of the previous season (sequential Bayesian updating), so the model 'remembers' recent seasonal trends (e.g., a run of below-average monsoons) while still being anchored to the long-run historical prior.
- Season-specific conditioning nodes: rainfall and temperature nodes are conditioned on the season/crop-cycle stage (sowing, vegetative, flowering, maturity), because the same rainfall deficit has very different impact on yield depending on which growth stage it occurs in — a Bayesian Network captures this by having separate conditional probability tables for rainfall-at-each-stage rather than a single seasonal-average rainfall node.

As the season progresses and more evidence is observed (actual rainfall received, updated soil moisture, mid-season temperature), the posterior over yield is repeatedly updated, so the prediction becomes progressively sharper (lower variance) as the season nears harvest — an important operational property, since early-season predictions are necessarily more uncertain than late-season ones and this should be communicated honestly to the end user.

3.5 Evaluating Model Reliability

Because the output is a probability distribution rather than a single number, reliability is evaluated along two dimensions: point-prediction accuracy and calibration of the uncertainty estimate.

Metric	What it Measures	Relevance
RMSE / MAE (posterior mean vs actual yield)	Point-prediction accuracy	Core accuracy check
Prediction Interval Coverage (e.g., 90% credible interval)	Whether the true yield falls inside the stated interval the expected % of the time	Calibration — are the uncertainty estimates trustworthy
Continuous Ranked Probability Score (CRPS)	Overall quality of the full predicted distribution vs the actual outcome	Rewards sharp AND well-calibrated forecasts
Posterior predictive checks	Whether simulated data from the model resembles real historical yield patterns	Model adequacy / goodness of fit

A reliable model should show narrowing credible intervals as the season progresses (reflecting genuinely reduced uncertainty), should have empirical interval coverage close to the nominal level (e.g., roughly 90% of seasons' true yields falling inside the stated 90% interval, not consistently higher

or lower), and should be periodically re-validated against extreme/outlier seasons, since climate variability means the model's stationarity assumptions can be violated by long-term climate shifts.

3.6 Impact on Decision-Making

For Farmers

- Input planning: probabilistic early-season forecasts inform decisions on fertiliser quantity, irrigation scheduling, and whether to switch to a drought-tolerant variety.
- Risk management: explicit uncertainty (not just a point forecast) supports decisions on crop insurance uptake and diversification.
- Market timing: a credible yield distribution supports better decisions on storage vs. immediate sale, and on contracting forward sales.

For Policy Planning

- Food security planning: aggregated regional yield-distribution forecasts help government agencies anticipate shortfall risk and plan buffer stock or import requirements ahead of harvest.
- Targeted support: districts whose posterior yield distribution shows a high probability of below-threshold yield can be prioritised for drought relief or subsidised input support.
- Insurance and subsidy design: calibrated probability distributions (rather than single-point estimates) are essential inputs to actuarially sound crop-insurance premium setting.

3.7 Conclusion

A Bayesian inference-based crop yield model allows prior agronomic knowledge and historical experience to be formally combined with real-time seasonal observations of rainfall, soil quality and temperature, producing not just a yield estimate but a calibrated distribution over likely outcomes that sharpens as the season progresses. This probabilistic framing is directly useful for both farm-level decisions (input use, insurance, sale timing) and policy-level planning (food security, targeted relief, insurance design), provided the model's calibration is continually validated and its priors are updated to reflect long-term climatic shifts rather than assumed to be static.